Gravity as Relational Difference Elimination: A Harmonic Closure Derivation

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Abstract

We present a geometric-informational model where gravity emerges as the macroscopic tendency to eliminate relational differences between projected states. The model is built upon the harmonic frequency partition $\omega_N^2 = \omega_V^2 + \omega_m^2$, previously validated in hadronic, atomic, and cosmological contexts within the same research program. The gravitational source is identified as the **informational content of confinement**: the relational charge density is $\rho_{\text{rel}} = \kappa\,\omega_m^2\,\rho_m$, where κ is the **specific relational volume** of the harmonic closure state. This leads to a Poisson equation $\nabla^2\phi = 4\pi\alpha\,\kappa\,\omega_m^2\,\rho_m$, recovering Newtonian gravity with $G = \alpha\,\kappa\,\omega_m^2$. The framework completes a **quadripartite unification**—proton radius, valence chemistry, Hubble expansion, and Newtonian gravity—spanning 46 orders of magnitude from a single geometric principle. We outline falsifiable avenues: composition-dependent violations of the weak equivalence principle, high-density / near-closure-scale corrections, and a scale-dependent effective coupling G_{eff} that can mimic dark-matter phenomenology under partial closure.

Keywords: emergent gravity; relational physics; harmonic partition; binary closure; unification

1 Introduction

Gravity remains the only fundamental interaction whose coupling constant G is imposed empirically rather than derived from deeper principles. While General Relativity provides an exquisite classical description, its conceptual origin—why mass curves spacetime with precisely this strength—remains unexplained.

In three independent domains, we have argued that a geometric framework based on relational projection and harmonic frequency partitions yields quantitative predictions without free parameters:

Table 1: Previous quantitative validations of the relational projection framework. For the proton, the "harmonic ratio" refers to the squared partition ratio $\omega_V^2 : \omega_m^2$ (see section 2.2).

Scale	**System**	**Harmonic ratio**	**Prediction / Result**
Hadronic	Proton charge radius	$\omega_V^2 : \omega_m^2 = 1 : 15$	$r_p^{(\text{model})} = 4\hbar/(m_p c) \approx 0.8412$ fm [1, 4]
Atomic	Valence stability (s–p)	$a : b$ prime or 2^n	Harmonic weight $P = a + b$ (classification rule) [2]
Cosmological	Hubble expansion	$V = \omega R$	$H_0 = \omega_V$ when $R = R_H$ (consistency) [3, 9]

This paper extends the same principle to gravity. We show that Newton's constant G is not fundamental but emerges as

$$G = \alpha\, \kappa\, \omega_m^2, \tag{1}$$

where ω_m is the confinement frequency (hadronic scale), κ is the specific relational volume of the harmonic closure state, and α is a universal dimensionless coupling. The framework does not predict the absolute numerical value of G without calibration—it *expresses* it in terms of microscopically meaningful quantities, shifting predictive power to variations and correlations when composition or closure state changes.

2 Relational Foundations

2.1 The Harmonic Partition Postulate

Postulate 1 (Universal Harmonic Partition). Every stable relational system admits an orthogonal decomposition of its total frequency ω_N:

$$\omega_N^2 = \omega_V^2 + \omega_m^2, \tag{2}$$

where ω_V is the projection frequency (visible, relational mode), ω_m is the confinement frequency (inertial, informational mode), and $\omega_N = mc^2/\hbar$ for massive systems. This is treated as a physical constraint selecting stable configurations through rational ratios $\omega_V : \omega_m$.

2.2 Binary Closure Principle

Postulate 2 (Binary Information Closure). A relational system achieves maximal stability when its harmonic partition satisfies a binary closure of depth n:

$$\frac{\omega_V^2}{\omega_N^2} = \frac{1}{2^n}, \qquad n \in \mathbb{N}. \tag{3}$$

From eq. (3) we obtain the equivalent forms:

$$\frac{\omega_m^2}{\omega_N^2} = 1 - \frac{1}{2^n} = \frac{2^n - 1}{2^n}, \tag{4a}$$

$$\omega_V^2 : \omega_m^2 = 1 : (2^n - 1), \tag{4b}$$

$$\omega_V : \omega_m = 1 : \sqrt{2^n - 1}. \tag{4c}$$

Observed ratios used in this program:

$$n = 2: \quad \omega_V^2 : \omega_m^2 = 1 : 3 \quad \Rightarrow \quad P = 4 = 2^2, \tag{5}$$

$$n = 4: \quad \omega_V^2 : \omega_m^2 = 1 : 15 \quad \Rightarrow \quad P = 16 = 2^4. \tag{6}$$

The integer n counts the depth of binary subdivision in the information manifold.

2.3 Identity Radius as Derived Quantity

The visible radius is defined operationally:

$$R_V \equiv \frac{c}{\omega_V}. \tag{7}$$

For a *nucleonic constituent* in its own closure state of depth n, eq. (3) implies $\omega_V = \omega_N/2^{n/2}$ with $\omega_N = m_p c^2/\hbar$. Therefore the nucleonic identity radius (per nucleon) is

$$R_{\rm id}^{(\rm nuc)} \equiv R_V = \frac{c}{\omega_V} = 2^{n/2}\,\bar{\lambda}_p, \qquad \bar{\lambda}_p = \frac{\hbar}{m_p c}. \tag{8}$$

Proton case ($n = 4$). Here $2^{n/2} = 4$ and

$$R_{\rm id}^{(\rm nuc)} = 4\bar{\lambda}_p = \frac{4\hbar}{m_p c} \approx 0.8412 \text{ fm}, \tag{9}$$

consistent with the CODATA-recommended proton charge radius within uncertainties [4].

Composite body scaling. For a macroscopic body consisting of N nucleons whose closure volume is extensive, the effective identity radius scales as

$$R_{\rm id} = N^{1/3}\,R_{\rm id}^{(\rm nuc)}. \tag{10}$$

In this paper $R_{\rm id}$ is treated as an *effective closure/identity scale* implied by the extensive closure-volume construction (not necessarily the geometric radius of a dilute macroscopic object).

3 Gravitational Model: Derivation from Harmonic Closure

3.1 Assumptions (Weak-Field Closure Regime)

(A1) Stationary, weak-field limit: no retardation, no radiation.

(A2) Euclidean 3-space and spherical symmetry outside sources.

(A3) Conservative potential field: $\mathbf{g} = -\nabla\phi$.

(A4) Outside sources, $\nabla^2\phi = 0 \Rightarrow$ unique $1/r$ solution $\Rightarrow \mathbf{g} \propto 1/r^2$.

(A5) Matching to Newton fixes only the normalization of ϕ in the classical limit.

3.2 What is the Gravitational Source?

Within this framework, mass is tied to confinement through ω_m. The simplest scalar source compatible with eq. (2) in the weak-field regime is taken as

$$\text{Source} \propto \omega_m^2\, \rho_m, \tag{11}$$

i.e., the informational content of confinement per unit volume.

3.3 Specific Relational Volume (Closure Cell)

For nucleonic matter in harmonic closure at binary depth n, define the closure-cell volume *per nucleon*:

$$v_{\text{id}} = \frac{4\pi}{3}\left(R_{\text{id}}^{(\text{nuc})}\right)^3 = \frac{4\pi}{3}\left(2^{n/2}\bar{\lambda}_p\right)^3. \tag{12}$$

For a composite system of N nucleons, the total closure volume is extensive:

$$V_{\text{cl}} = N v_{\text{id}}. \tag{13}$$

The specific relational volume (volume per unit mass) is therefore

$$\kappa \equiv \frac{V_{\text{cl}}}{M} = \frac{N v_{\text{id}}}{N m_p} = \frac{v_{\text{id}}}{m_p}. \tag{14}$$

Crucially, κ depends on closure state (via n) and the nucleon mass m_p, but **not on system size** N. This preserves $Q \propto M$ and Newtonian scaling.

3.4 Relational Charge and Poisson Equation

Define the relational charge density

$$\rho_{\text{rel}} \equiv \kappa\, \omega_m^2\, \rho_m, \tag{15}$$

where ρ_m is ordinary mass density. Introduce a dimensionless coupling α. The gravitational potential satisfies

$$\nabla^2\phi = 4\pi\alpha\, \rho_{\text{rel}} = 4\pi\alpha\, \kappa\, \omega_m^2\, \rho_m. \tag{16}$$

Dimensional check. $\nabla^2\phi \sim [T]^{-2}$, $\rho_m \sim [M][L]^{-3}$, $\kappa \sim [L]^3[M]^{-1}$, $\omega_m^2 \sim [T]^{-2}$, $\alpha \sim 1$. Hence the right-hand side of eq. (16) has dimension $[T]^{-2}$, consistent.

3.5 Newton's Constant as a Derived Quantity

For a point mass M, the solution to eq. (16) is

$$\phi(r) = -\frac{\alpha\,\kappa\,\omega_m^2\,M}{r}. \tag{17}$$

Comparing with Newton's $\phi_N(r) = -GM/r$ yields

$$\boxed{G = \alpha\,\kappa\,\omega_m^2}, \tag{18}$$

which matches eq. (1). This is **not** a prediction of the absolute numerical value of G without calibration; rather, it expresses G in terms of (i) a confinement scale ω_m, (ii) a closure geometry scale κ, and (iii) a universal dimensionless coupling α.

3.6 Numerical Calibration and Consistency

For nucleonic matter in ground-state closure ($n = 4$), one has $R_{\rm id}^{(\rm nuc)} = 4\bar{\lambda}_p$ and

$$\kappa = \frac{4\pi}{3}\frac{(4\bar{\lambda}_p)^3}{m_p} \approx 1.49 \times 10^{-18}\ \mathrm{m^3\,kg^{-1}}, \tag{19}$$

$$\omega_m^2 = \left(\frac{\sqrt{15}}{4}\omega_p\right)^2 = \frac{15}{16}\omega_p^2, \qquad \omega_p = \frac{m_p c^2}{\hbar} \approx 1.4255 \times 10^{24}\ \mathrm{s^{-1}},$$
$$\omega_m^2 \approx 1.905 \times 10^{48}\ \mathrm{s^{-2}}. \tag{20}$$

Therefore

$$G = \alpha\,\kappa\,\omega_m^2 \approx \alpha\,(2.84 \times 10^{30}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}). \tag{21}$$

Setting $G = 6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ (CODATA 2018 [4]) gives

$$\alpha \approx 2.35 \times 10^{-41}. \tag{22}$$

This extremely small value reflects the weakness of gravity relative to the confinement scale. Once calibrated, α is assumed universal.

4 Unification Across Scales

5 Testable Predictions and Falsifiability

5.1 Composition Dependence (Weak Equivalence Principle)

To make the weak-equivalence discussion explicit, we adopt the minimal interaction-energy ansatz that a test body carries a *relational charge* proportional to its confinement content,

$$q_{\rm rel} \equiv \kappa_t\,\omega_{m,t}^2\,m, \tag{23}$$

Table 2: Quadripartite unification via $\omega_N^2 = \omega_V^2 + \omega_m^2$.

Scale	**System**	n	ω_V	ω_m	$R_{\mathbf{id}}$	**Emergent property**
Hadronic	Proton	4	$\omega_p/4$	$\sqrt{15}\omega_p/4$	$4\bar{\lambda}_p$	Charge radius
Atomic	s–p valence	1,2	N_s freq.	N_p freq.	Bohr scale	Harmonic weight P
Cosmological	Hubble flow	0 (limit)	H_0	$\to 0$	$R_H = c/H_0$	Expansion law
Gravitational	Newtonian	4 (nucleonic)	$c/R_{\rm id}$	ω_m	$(3\kappa M/4\pi)^{1/3}$	$G = \alpha\kappa\omega_m^2$

where m is the inertial mass of the test body and $(\kappa_t, \omega_{m,t})$ characterize its effective closure state (coarse-grained over constituents). The potential energy is $U = \alpha\, q_{\rm rel}\, \phi$, so $\mathbf{F} = -\nabla U$ gives

$$\mathbf{a} = \frac{\mathbf{F}}{m} = -\alpha \left(\frac{q_{\rm rel}}{m}\right) \nabla\phi = -\alpha\, \kappa_t\, \omega_{m,t}^2\, \nabla\phi. \tag{24}$$

Thus WEP holds exactly only when $\kappa_t\, \omega_{m,t}^2$ is universal across test bodies. If $\omega_{m,t}$ (or κ_t) varies slightly with composition (e.g., via binding energy per nucleon), the model predicts small violations.

Define the Eötvös parameter

$$\eta_{A,B} \equiv 2\frac{|a_A - a_B|}{a_A + a_B} \approx \frac{\Delta(\kappa_t\, \omega_{m,t}^2)}{\kappa_t\, \omega_{m,t}^2} \approx \frac{\Delta(\omega_m^2)}{\omega_m^2} + \frac{\Delta\kappa}{\kappa}, \tag{25}$$

where the last step assumes small fractional differences. **Prediction:** η should correlate with mean binding energy per nucleon, not just atomic mass. Current limits from MICROSCOPE [8] give $\eta \lesssim 10^{-15}$. A more quantitative estimate requires an explicit microphysical map from nuclear composition to ω_m within the program.

5.2 High-Density / Near-Closure-Scale Corrections

When $r \lesssim R_{\rm id}$, the point-source approximation fails. The potential may acquire higher-order corrections from internal closure structure:

$$\phi(r) = -\frac{GM}{r}\left[1 + \beta\left(\frac{R_{\rm id}}{r}\right)^2 + \gamma\left(\frac{R_{\rm id}}{r}\right)^4 + \cdots\right]. \tag{26}$$

For neutron stars ($N \sim 10^{57}$), eq. (10) gives $R_{\rm id} \sim N^{1/3} R_{\rm id}^{(\rm nuc)} \sim 10^4$ m. Hence for exterior fields at $r \gg R_{\rm id}$ the corrections are suppressed as $\mathcal{O}((R_{\rm id}/r)^2)$, while the interior regime $r \lesssim R_{\rm id}$ provides a falsifiable strong-density domain once a relativistic completion is specified.

5.3 Scale-Dependent Gravity and Dark-Matter Phenomenology

In diffuse systems (e.g., galaxies), the effective closure state may differ from the ground state. If closure is partial, the effective specific volume $\kappa(r)$ becomes scale-dependent. Define

$$G_{\rm eff}(r) \equiv \alpha\, \kappa(r)\, \omega_m^2. \tag{27}$$

A minimal phenomenological ansatz for a radially varying closure depth is

$$\kappa(r) = \kappa_0 \left[1 + \lambda \left(\frac{r}{r_0} \right)^{\delta} \right], \tag{28}$$

which can produce flat rotation curves for suitable λ, δ. **Prediction:** rotation-curve anomalies should correlate with the informational density $\omega_m^2 \rho_m$, not only with mass distribution. This provides a concrete, falsifiable signature distinguishing the present framework from both ΛCDM and MOND [11, 12].

6 Discussion

6.1 What Has Been Achieved

We have transformed gravity from:

- empirical parameter → derived form $G = \alpha\kappa\omega_m^2$;
- potential circularity → geometric consequence via $\kappa = v_{\text{id}}/m_p$ and $R_{\text{id}} = (3\kappa M/4\pi)^{1/3}$;
- isolated force → fourth manifestation of harmonic relational closure.

6.2 Cosmological Consistency (Assumption-Dependent Check)

One may perform an *internal consistency* check by inserting an order-of-magnitude estimate for the mass scale of the observable universe into the closure-volume scaling. Because the total mass budget depends on cosmological parameters and modeling choices, this step is not presented as a standalone falsifiable prediction; it is an assumption-dependent plausibility estimate.

If one takes $M_U \sim 10^{53}$ kg (order-of-magnitude consistent with standard cosmological parameters [9]) and uses the calibrated α, the closure depth estimate obtained from eq. (1) and eq. (14) is

$$n \approx \left(\frac{3GM_U}{4\pi\alpha\,\omega_m^2\,\bar{\lambda}_p^3} \right)^{1/3} \sim 10^{13}\ (\approx 2^{45}), \tag{29}$$

which is not unreasonable within a hierarchical closure picture. The utility of eq. (29) is primarily to constrain internal parameter consistency once the program specifies the relevant cosmic-scale closure assumptions.

6.3 Relation to Other Approaches

6.4 Limitations and Future Work

Limitations:

- Weak-field, stationary regime only.
- Relativistic completion (e.g., coupling to a projection tensor $P_{\mu\nu}$ [3]) remains future work.

Table 3: Comparison with other emergent-gravity frameworks.

Approach	**Source of Gravity**	**Status of G**	**Unification**
Newton/Einstein	Mass/stress-energy	Fundamental constant	None
Verlinde (entropic) [6]	Information entropy	Emergent (qualitative)	Qualitative
Jacobson (thermodynamic) [5]	Horizon thermodynamics	Proportionality	Qualitative
Padmanabhan (thermo-gravity) [7]	Horizon/thermodynamics	Emergent (review)	Qualitative
This work	Harmonic closure volume	**Derived:** $G = \alpha\kappa\omega_m^2$	**Quantitative** (46 orders)

- α is calibrated, not derived from first principles.
- The test-body coupling in eq. (23)–eq. (24) is a minimal phenomenological closure to make WEP tests explicit; a full microphysical derivation remains open.

Future directions:

- Derive α from a higher-order harmonic closure condition.
- Develop a strong-field tensor formulation (black holes, neutron stars).
- Quantitatively fit galaxy rotation curves using eq. (28).

7 Conclusion

We have presented a complete, non-circular, dimensionally consistent derivation of Newtonian gravity from relational projection principles. The key elements are:

1. Relational charge density $\rho_{\text{rel}} = \kappa\,\omega_m^2\,\rho_m$, with $\kappa = v_{\text{id}}/m_p$ the specific relational volume of the harmonic closure state.
2. Poisson equation $\nabla^2\phi = 4\pi\alpha\,\kappa\,\omega_m^2\,\rho_m$ yielding Newtonian limit with $G = \alpha\,\kappa\,\omega_m^2$.
3. κ determined geometrically from binary depth n: $\kappa = \dfrac{4\pi}{3}(2^{n/2}\bar{\lambda}_p)^3/m_p$.
4. Identity radius $R_{\text{id}} = (3\kappa M/4\pi)^{1/3}$ emerges naturally, resolving any circularity.
5. Falsifiable avenues: composition-dependent WEP deviations (via test-body relational charge), near-closure-scale corrections, and scale-dependent G mimicking dark matter under partial closure.
6. Quadripartite unification across 46 orders of magnitude: proton radius, valence chemistry, Hubble expansion, and Newtonian gravity—all from $\omega_N^2 = \omega_V^2 + \omega_m^2$ with binary closure.

Gravity, like the size of the proton, the stability of atoms, and the expansion of the cosmos, is **not a fundamental interaction** but a **manifestation of relational geometry seeking harmonic closure**.

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